

Assignment 13

Module -1 - Linux server - Understand and use essential tools

1. What is the minimum number of partitions you need to install Linux?

❖ Minimum 2 partitions are generally recommended:

1. Root Partition :

- Stores Linux operating system files.

2. Swap Partition :

- Used as virtual memory when RAM becomes full.

2. Explain About Chmod Command

➤ The chmod command is used to change file and directory permissions in Linux.

❖ Syntax

- chmod permissions filename

3. How to check Linux memory utilization

❖ Command

- free -h

❖ **Why it is Used?**

It shows:

- Total RAM
- Used memory
- Free memory
- Swap memory

4. Use grep to search for specific patterns in files.

❖ Command

- grep "word" filename

❖ **Why grep is Important?**

➤ It searches text quickly inside files.

Useful for:

- Log analysis
- Error searching
- Finding configuration settings

5. Get Connecting on a linux server by ssh

❖ Command

- ssh username@ipaddress

❖ Why SSH is Used?

- Secure Shell provides secure remote access to Linux servers.

Benefits:

- Encrypted communication
- Remote administration
- Secure file transfer

6. Create 5 files in the /tmp directory, and then use tar and gzip to bundle and compress the files.

step1: create files

- touch /tmp/file1 /tmp/file2 /tmp/file3 /tmp/file4 /tmp/file5

step2: create tar file

- tar -cvf files.tar /tmp/file1 /tmp/file2 /tmp/file3 /tmp/file4 /tmp/file5

step3: compress using gzip

- gzip files.tar

❖ Why tar and gzip are Used?

- tar bundles files together.
- gzip reduces file size.

Useful for:

- Backup
- File transfer
- Storage saving

7. Describe the root account

- ❖ The root account is the administrator account in Linux.

Features

- Full system access
- Can modify any file
- Can install/remove software
- Can manage users

❖ Why is root Important?

It is required for:

- System administration
- Configuration changes
- Security management

8. What is shell?

- ❖ A shell is a command-line interface between the user and the Linux kernel.

Functions

- Accepts user commands
- Executes programs
- Manages scripts

❖ Why shell is important:

- It allows users to interact with the operating system.

9. What is Linux?

- Linux is an open-source operating system based on Unix.

❖ Features

- Multi-user
- Multitasking
- Secure
- Stable

❖ Why Linux is Popular?

- Free to use
- Highly secure
- Used in servers and cloud computing

10. What is Bash?

- Bash stands for **Bourne Again Shell**
- It is the default shell in many Linux distributions.

❖ Functions

- Executes commands
- Runs shell scripts
- Automates tasks

11. You have a new empty hard drive that you will use for Linux. What is the first step you use.

➤ First Step

- Create partitions using tools like:
 - fdisk
 - or
 - parted

❖ Why:

- Linux needs partitions to:
- Store OS files
- Manage data
- Organize storage

12. Write the Linux command to show the current working directory.

❖ Command

- pwd

❖ Meaning

- pwd = Present Working Directory

13. write the Linux command to get help with various options.

❖ Command

- man commandname

❖ Why?

Displays:

- Command usage
- Options
- Examples

14. Write the linux command to display what all users are currently doing.

❖ Command

- w

❖ Why?

Shows:

- Logged-in users
- Running processes
- System activity

15. write the Linux command to get information about the operating system.

❖ Command

- uname -a

❖ Why?

Displays:

- Kernel version
- OS architecture
- System details

16. Write the Linux command to create a hard link of a file.

❖ Command

- ln originalfile hardlink

❖ Why:

- Creates another reference to the same file data.

17. Write the Linux command to create a soft link of a file as well as Directory.

❖ Command

- `ln -s source destination`

❖ Why :

- Soft links work like shortcuts.

18. Write the Linux command! to search for specific pattern in a file.

❖ Command

- `grep "pattern" filename`

❖ Why :

- Used for quick text searching.

19. Write the Linux command to show the use of basic regular expressions using grep command.

❖ Command

- `grep "^a" file.txt`

➤ Searches lines starting with letter a.